Appendix: Bill of Materials

Table A lists the bought-out electronics and power parts, Table B lists every mechanical part of the CAD assembly with its mass, and Table C gives the mass by sub-assembly.

A. Electronics and Power Components (bought-out)

Qty	Component	Make / Company	Used for
1	Raspberry Pi 5, 8 GB, with Active Cooler	Raspberry Pi	Vision and state machine
1	Raspberry Pi AI HAT+ (Hailo-8L, 13 TOPS)	Raspberry Pi	Runs the YOLOv8n detector
2	Raspberry Pi Camera Module 3 Wide (120 deg)	Raspberry Pi	Stereo cameras
1	Black Pill STM32F411	WeAct Studio	Real time control and safety
1	BNO085 IMU module	Generic	Body orientation
1	INA226 current and power monitor module	Generic	Pack current and over current trip
3	MP1584EN buck converter module	Generic	5.1 V, 5.0 V and 6.0 V rails
1	12 V buck module, 5 A or more, 40 V input or more	Generic	Steering driver and motor supply
1	Voltage clamp board (100 V N-MOSFET, 27 V zener, 10 k resistor)	In-house	Protects the logic buck inputs
1	SMBJ28A TVS diode	Generic	Logic branch spike protection
1	RXB6 433 MHz receiver	Generic	Wireless kill switch
1	PT2272-L4 latched decoder	Generic	Kill switch latch
1	433 MHz key fob (STOP / RUN)	Generic	Wireless kill switch transmitter
1	Cytron MDD10A dual motor driver	Cytron	Drive motors
1	Cytron MD10C R3 motor driver	Cytron	Steering motor
2	775 DC motor, 24 V, with PG45 19:1 gearbox and hall encoder	Generic	Drive
1	JGY-370 worm gearmotor, 12 V, with encoder	Generic	Steering
1	MG996R metal gear servo	Generic	Camera gimbal
2	LiPo 6S 5000 mAh	Turnigy nano-tech	Main battery, two in parallel
1	DC contactor, 24 V coil, 100 A	Generic	Motor power switching
1	MIDI fuse holder with 60 A fuse	Generic	Motor branch protection
1	5 A blade fuse with holder	Generic	Logic branch protection
1	XT90-S anti-spark port with XT90 loop key	Amass	Arming key and manual disconnect
1	XT90 charge port with dust cap	Amass	Off-robot charging
2	Electrolytic capacitor, 470 uF, 35 V, low ESR	Generic	At the driver inputs
1	DHVAJA Main Control Board (KiCad, 2 layer, 120 x 108 mm)	In-house	Hosts all logic parts

B. Mechanical and Structural Parts (from the CAD assembly)

#	Item	Qty	Material / Type	Mass each (g)	Mass total (g)
Shell					
1	Shell hemisphere, PC 3 mm, OD 640	2	Polycarbonate	2236.0	4471.9
2	Equator T-ring (rolled + machined)	1	Al 6061-T6	606.5	606.5
3	Tread band, TPU 95A, 70 x 5 mm	1	TPU 95A	817.3	817.3
Hubs					
4	Hub carrier (CNC)	2	Al 6061-T6	996.4	1992.9
5	Hub cover	2	Al 6061-T6	259.0	517.9
6	Internal ring gear 72T m2	2	Al 7075-T6	303.5	607.0
7	Bearing 6806-2RS (30x42x7)	4	Steel	20.9	83.5
8	Circlip DIN 472-42	2	Steel	8.2	16.4
Frame					
9	Main shaft, 7075 tube 30x3	1	Al 7075-T6	451.3	451.3
10	Circlip DIN 471-30	2	Steel	4.3	8.6
11	Shaft clamp flange (split)	2	Al 6061-T6	93.0	186.0
12	Side plate, 6 mm	2	Al 6061-T6	675.0	1350.1
13	Rubber grommet 16 mm	4	TPU 95A	1.3	5.0
20	Bulkhead, 6 mm	2	Al 6061-T6	253.9	507.8
21	Electronics deck, CF 3 mm	1	Carbon fibre	237.8	237.8
Drive					
14	PG45 planetary gearbox 19:1	2	Bought-out	360.0	720.0
15	775 DC motor 24 V	2	Bought-out	380.0	760.0
16	Hall encoder (on motor)	2	Bought-out	60.0	120.0
17	Motor face bracket, 6 mm	2	Al 6061-T6	95.9	191.8
18	Pinion 18T m2, D-bore 12	2	Steel	105.7	211.4
19	Hex spacer M5 x 124	8	Al 6061-T6	24.4	194.8
Steering					
22	Steering bearing housing	2	Al 6061-T6	29.1	58.3
23	Bearing 6001-2RS (12x28x8)	2	Steel	17.7	35.4
24	Steering shaft 12 mm EN8	1	Steel	222.7	222.7
25	Circlip DIN 471-12	2	Steel	0.5	0.9
26	Rigid coupling 6/12	1	Al 6061-T6	13.3	13.3
27	JGY-370 worm gearmotor 12 V (gearbox)	1	Bought-out	170.0	170.0
28	JGY-370 motor + encoder	1	Bought-out	90.0	90.0
29	Steering motor bracket	1	Al 6061-T6	25.7	25.7
30	Round spacer M4 x 39	4	Al 6061-T6	4.4	17.6
Pendulum					
31	Pendulum arm, 6 mm	2	Al 6061-T6	195.7	391.4
32	Battery tray, 6 mm	1	Al 6061-T6	866.5	866.5
33	Ballast plate, mild steel 8 mm	5	Steel	1072.3	5361.5
34	Battery cradle (PETG print)	1	PETG (3D print)	113.2	113.2
Power					
35	LiPo 6S 5000 mAh (Turnigy nano-tech)	2	Bought-out	830.0	1660.0
36	Battery strap	2	Bought-out	10.0	20.0
37	DC contactor 24 V 100 A	1	Bought-out	280.0	280.0
38	MIDI fuse holder + 60 A fuse	1	Bought-out	45.0	45.0
77	XT90-S anti-spark port (arming) + XT90 loop key	1	Bought-out	22.0	22
78	XT90 charge port + dust cap	1	Bought-out	16.0	16

#	Item	Qty	Material / Type	Mass each (g)	Mass total (g)
Vision					
39	Camera gimbal tube, CF 20 mm	1	Carbon fibre	37.8	37.8
40	Flanged bearing F6804-2RS	2	Steel	30.7	61.4
41	Shaft collar 20 mm	2	Al 6061-T6	3.1	6.2
42	Servo MG996R	1	Bought-out	55.0	55.0
43	Servo coupler (PETG print)	1	PETG (3D print)	4.2	4.2
44	Servo bracket (PETG print)	1	PETG (3D print)	20.7	20.7
45	Camera housing (PETG print)	2	PETG (3D print)	14.0	28.0
73	Raspberry Pi Camera Module 3 Wide (120 deg)	2	Bought-out	4.0	8
Electronics					
71	Raspberry Pi 5 8 GB + Active Cooler	1	Bought-out	66.0	66
72	Raspberry Pi AI HAT+ (Hailo-8L, 13 TOPS)	1	Bought-out	25.0	25
74	Cytron MDD10A dual motor driver (drive)	1	Bought-out	70.0	70
75	Cytron MD10C R3 motor driver (steering)	1	Bought-out	30.0	30
PCB					
76	DHVAJA Main Control Board (KiCad, 2-layer, 120 x 108)	1	Bought-out	190.0	190
Fasteners					
46	Countersunk screw ISO 10642 M4x8	32	Steel 12.9 / A2	1.1	33.7
47	Socket head cap screw ISO 4762 M4x25	16	Steel 12.9 / A2	3.3	52.6
48	Button head screw ISO 7380 M5x16	16	Steel 12.9 / A2	3.2	51.7
49	Socket head cap screw ISO 4762 M5x20	4	Steel 12.9 / A2	3.4	13.8
50	Socket head cap screw ISO 4762 M4x12	12	Steel 12.9 / A2	2.2	27.0
51	Grub screw ISO 4026 M4x6	4	Steel 12.9 / A2	0.5	2.0
52	Socket head cap screw ISO 4762 M5x12	24	Steel 12.9 / A2	3.8	90.0
53	Socket head cap screw ISO 4762 M4x16	14	Steel 12.9 / A2	2.6	37.0
54	Countersunk screw ISO 10642 M3x10	8	Steel 12.9 / A2	0.7	5.3
55	Socket head cap screw ISO 4762 M3x14	6	Steel 12.9 / A2	1.3	7.6
56	Socket head cap screw ISO 4762 M3x6	2	Steel 12.9 / A2	0.8	1.6
57	Socket head cap screw ISO 4762 M4x10	4	Steel 12.9 / A2	2.0	8.2
58	Countersunk screw ISO 10642 M4x16	6	Steel 12.9 / A2	1.8	11.0
59	Countersunk screw ISO 10642 M6x28	4	Steel 12.9 / A2	7.1	28.3
60	Countersunk screw ISO 10642 M6x22	4	Steel 12.9 / A2	5.7	23.0
61	Button head screw ISO 7380 M4x30	4	Steel 12.9 / A2	3.3	13.4
62	Socket head cap screw ISO 4762 M5x10	2	Steel 12.9 / A2	3.4	6.9
63	Grub screw ISO 4026 M3x5	2	Steel 12.9 / A2	0.2	0.5
64	Socket head cap screw ISO 4762 M3x12	8	Steel 12.9 / A2	1.0	8.1
65	Socket head cap screw ISO 4762 M3x10	8	Steel 12.9 / A2	1.0	7.7
66	Socket head cap screw ISO 4762 M3x16	4	Steel 12.9 / A2	1.1	4.5
67	MCB Standoffs M3x10	12	Steel 12.9 / A2	1.1	13.2
68	Pi Standoffs M2.5x6	12	Steel 12.9 / A2	0.6	7.5
69	MDD10A Standoffs M3x8	12	Steel 12.9 / A2	1.0	11.9
70	MD10C Standoffs M3x8	12	Steel 12.9 / A2	1.0	11.9

C. Mass by Sub-assembly

Sub-assembly	Mass (g)	Share
Shell	5896	24.0 %
Hubs	3218	13.1 %
Frame	2747	11.2 %
Drive	2198	9.0 %
Steering	634	2.6 %
Pendulum	6733	27.4 %
Power	2043	8.3 %
Vision	221	0.9 %
Electronics	191	0.8 %
PCB	190	0.8 %
Fasteners	478	1.9 %
Total of the table	24548	100 %

The table adds up to 24.55 kg. The CAD mass properties used in Section 1.4 give 24.80 kg, the remaining 252 g being small electronics and wiring that are not listed as separate rows.